

Dependency-Colored Concentration for Finite-Alphabet Shared-Exclusions PID

Finite-sample law control, drift separation, local atom moduli,
and a formal evidence boundary

pid-rs project analysis

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Abstract

This document gives new project-defined validation for the Wibral-lineage categorical shared-exclusions partial information decomposition (SxPID) in `pid-rs`. The paper-defined functional is not new. The document derives a finite-sample concentration bound for the complete empirical joint law when all complete rows share one common finite law and a fixed coloring partitions observations into classes whose complete rows are mutually independent. Dependence across classes can be arbitrary. A class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme improves the ordinary color-count factor for unbalanced classes. A telescoping error allocation gives a time-uniform envelope. A nonidentical-law extension separates random error from explicit drift. A stronger common-support modulus then transfers joint-law error to pointwise and averaged SxPID atoms. Lean checks the deterministic core, and an executable challenge suite retains counterexamples to weaker assumptions. The result does not validate continuous shared exclusions, arbitrary time series, adaptive evaluation transforms, or binary64 asymptotics.

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1 Status, scope, and provenance

The mathematical functional, the concentration ingredients, and this repository artifact have different origins.

Object	Origin	Status in this document
Categorical SxPID	Paper-defined by Makkeh, Gutknecht, and Wibral; its lattice has a formal-logic basis due to Gutknecht, Wibral, and Makkeh [1, 2]	Implemented for two to four sources; receives a finite-sample bound and an almost-sure exact-real consistency implication when the displayed envelope vanishes
Hoeffding bounded-variable inequality	Paper-defined classical result [3]	Used inside each independent color class
Dependency-color concentration	Published background includes Janson’s coloring method and later Hölder formulations [4, 5]	Project-defined proof composition with a class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme; no novelty claim
Empirical-law L^1 deviation	Published independent and identically distributed (i.i.d.) background includes Weissman et al. [6]	One-sided subset union bound under the stated color contract
SxPID local modulus	Project-defined validation of the paper-defined functional	Safe exact-real common-support baselines with exact Möbius row norms; no sharpness claim
Drift envelope	Project-defined extension	Concentration about the average row law plus explicit bias to a reference law

The machine-readable authority is `method-catalog.json`. The rendered authority is `METHODS.md`. The Rust implementation routes covered by the final composition are the stable categorical functions

`discrete_sxpid2`, `discrete_sxpid2_averaged`, `discrete_sxpid3`,
`discrete_sxpid3_averaged`, `discrete_sxpid_n`, and `discrete_sxpid_n_averaged`.

They are exported from `stable::categorical`. This document adds no estimator and no public Rust API. Its proof and executable artifacts are validation evidence.

2 Finite setting and the dependence contract

Let $Z_1, \dots, Z_n$ take values in a finite alphabet $\mathcal{Z}$ with $|\mathcal{Z}| = K$. For SxPID, one row is the complete joint symbol

$$Z_i = (S_{1i}, \dots, S_{mi}, T_i). \quad (1)$$

Dependence between source and target coordinates inside a row is unrestricted. It is the signal that PID analyzes.

For the common-law theorem, assume every row has marginal law P . Let a deterministic color map partition the indices into occupied classes $C_1, \dots, C_r$. Define

$$n_j = |C_j| > 0, \quad n = \sum_{j=1}^r n_j. \quad (2)$$

The complete rows $\{Z_i : i \in C_j\}$ must be mutually independent for every j . Dependence across different classes can be arbitrary. The color map must not depend on the observed row values.

Define

$$\hat{P}_n(z) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Z_i = z\}, \quad V_n = \left(\sum_{j=1}^r \sqrt{n_j} \right)^2, \quad d_{\text{eff},n} = \frac{V_n}{n}. \quad (3)$$

Proposition 2.1 (Effective-color range). *The deterministic proxy obeys*

$$n \leq V_n \leq rn, \quad 1 \leq d_{\text{eff},n} \leq r. \quad (4)$$

Proof. For the lower bound, expand the square of a sum of nonnegative square roots and retain its diagonal terms. For the upper bound, apply Cauchy–Schwarz:

$$\left(\sum_j \sqrt{n_j} \right)^2 \leq \left(\sum_j 1^2 \right) \left(\sum_j n_j \right) = rn. \quad (5)$$

The Lean artifact proves both inequalities over exact real numbers. $\square$

The value $d_{\text{eff},n}$ is a proof constant derived from the declared class sizes. It is not an estimated effective sample size.

3 Finite-sample law concentration

Theorem 3.1 (Dependency-colored empirical-law bound). *Assume $K \geq 2$ and the contract in Section 2. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\hat{P}_n - P\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, (2^K - 2) \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (6)$$

If at most d colors are occupied, then

$$\mathbb{P}\left(\left\|\hat{P}_n - P\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, (2^K - 2) \exp\left(-\frac{n \varepsilon^2}{2d}\right)\right\}. \quad (7)$$

Proof. Fix a nonempty proper subset $A \subset \mathcal{Z}$. Put

$$Y_i = \mathbf{1}\{Z_i \in A\} - P(A), \quad S_j = \sum_{i \in C_j} Y_i. \quad (8)$$

Let $R = \sum_j \sqrt{n_j}$ and select the Hölder exponents

$$p_j = \frac{R}{\sqrt{n_j}}, \quad \sum_j \frac{1}{p_j} = 1. \quad (9)$$

Each $p_j \geq 1$ because $R \geq \sqrt{n_j}$, so these are valid Hölder exponents. For $\lambda > 0$, generalized Hölder gives

$$\mathbb{E} \exp \left(\lambda \sum_j S_j \right) \leq \prod_j \left(\mathbb{E} e^{p_j \lambda S_j} \right)^{1/p_j}. \quad (10)$$

Rows are mutually independent inside each color. Hoeffding's lemma for a centered random variable with range length one gives

$$\mathbb{E} e^{p_j \lambda S_j} \leq \exp \left(\frac{n_j p_j^2 \lambda^2}{8} \right). \quad (11)$$

Therefore,

$$\mathbb{E} \exp \left(\lambda \sum_j S_j \right) \leq \exp \left(\frac{\lambda^2 V_n}{8} \right). \quad (12)$$

Chernoff's method, optimized at $\lambda = 4nt/V_n$, yields

$$\mathbb{P} \left(\hat{P}_n(A) - P(A) \geq t \right) \leq \exp \left(-\frac{2n^2 t^2}{V_n} \right). \quad (13)$$

For any signed finite measure with total mass zero,

$$\frac{1}{2} \left\| \hat{P}_n - P \right\|_1 = \max_{A \subseteq \mathcal{Z}} \{ \hat{P}_n(A) - P(A) \}. \quad (14)$$

Set $t = \varepsilon/2$ and take a union bound over the $2^K - 2$ nonempty proper subsets. This proves Equation (6). Complement subsets already encode the opposite sign, so no additional factor of two is required. Proposition 2.1 gives Equation (7). $\square$

For $K = 1$, the L^1 distance is identically zero, and the finite-sample formulas reduce to the true but empty statement $0 \leq 0$ for every $\varepsilon > 0$. The time-uniform radius contains $\log(2^K - 2)$, so Section 4 requires $K \geq 2$. The subset factor is exponential in K , so the bound can be vacuous on a large alphabet. This result makes no sharpness or sample-complexity claim. If $s = |\text{supp}(P)| \geq 2$ is known before the data are inspected, then $2^s - 2$ can replace $2^K - 2$. An observed support size is not a known population support size. When $s = 1$, the law distance is zero and the result is vacuous.

Proposition 3.2 (Hölder proxy optimum). *For $q_j > 0$ with $\sum_j q_j^{-1} = 1$,*

$$V_n \leq \sum_j n_j q_j. \quad (15)$$

Equality holds for $q_j = R/\sqrt{n_j}$.

Proof. The exponent conditions force $q_j \geq 1$ for every j . Thus, they are valid Hölder exponents. Cauchy–Schwarz gives

$$\left(\sum_j \sqrt{n_j} \right)^2 = \left(\sum_j \sqrt{n_j q_j} \sqrt{q_j^{-1}} \right)^2 \leq \left(\sum_j n_j q_j \right) \left(\sum_j q_j^{-1} \right). \quad (16)$$

The selected exponents make the two Cauchy–Schwarz vectors proportional. $\square$

This is an optimum inside the Hölder–Hoeffding proof scheme for this partition. It is not a claim that V_n is the best possible constant for every admissible joint law.

4 Time-uniform envelope and consistency

Assume $K \geq 2$. For an infinite sequence, require one fixed map from the positive integers to a finite or countable color set. Only finitely many colors can be occupied in each prefix. The entire infinite collection in each color must consist of mutually independent complete rows. Let V_n use the nonempty class sizes in the first n rows. A uniform bound of at most d occupied colors is required only for the coarse radius in Equation (23).

Fix $0 < \alpha < 1$ and define

$$C_K = 2^K - 2, \quad \alpha_n = \frac{\alpha}{n(n+1)}. \quad (17)$$

The allocation has the exact finite identity

$$\sum_{n=1}^N \alpha_n = \alpha \left(1 - \frac{1}{N+1}\right) \leq \alpha. \quad (18)$$

Lean proves the unit-scale finite identity by induction. Multiplication by the fixed scalar α gives Equation (18).

Define

$$R_n = \sqrt{\frac{2V_n}{n^2} \log\left(\frac{C_K n(n+1)}{\alpha}\right)}, \quad (19)$$

$$D_n = \min\{2, R_n\}. \quad (20)$$

Theorem 4.1 (Anytime law envelope). *Under the infinite-sequence contract,*

$$\mathbb{P}(\forall n \geq 1 : \|\hat{P}_n - P\|_1 \leq D_n) \geq 1 - \alpha. \quad (21)$$

Proof. For every n , substitution in Theorem 3.1 gives

$$C_K \exp\left(-\frac{n^2 R_n^2}{2V_n}\right) = \alpha_n. \quad (22)$$

The Lean artifact checks the algebraic exponent cancellation. Positivity, the square-root substitution, and its probability interpretation remain in this prose proof. A union bound and Equation (18) give simultaneous control by R_n . Every probability-law L^1 distance is also at most two. Thus it is at most $D_n = \min\{2, R_n\}$. $\square$

The coarser d -color radius is

$$D_n^{(d)} = \min\left\{2, \sqrt{\frac{2d}{n} \log\left(\frac{C_K n(n+1)}{\alpha}\right)}\right\}. \quad (23)$$

Under one uniform fixed bound d , this is $O(\sqrt{d \log(n)/n})$. For each positive rational error threshold, the tails in Equation (7) are summable. Borel–Cantelli and a countable intersection over those thresholds therefore give

$$\|\hat{P}_n - P\|_1 \longrightarrow 0 \quad \text{almost surely.} \quad (24)$$

If the number of colors grows, the displayed envelope converges to zero under the sufficient condition

$$\frac{V_n \log n}{n^2} \longrightarrow 0. \quad (25)$$

This condition is not necessary for empirical-law convergence. For example, i.i.d. rows still converge when a needlessly fine singleton coloring gives $V_n = n^2$. The condition is sufficient for this declared-color envelope. A singleton coloring also permits complete dependence across rows, so the theorem cannot infer convergence from that coloring alone.

Corollary 4.2 (Fixed-width overlapping windows). *Let E_t be i.i.d. innovations. Fix $w \in \mathbb{N}$ before evaluation, and let*

$$Z_t = f(E_t, \dots, E_{t+w-1}) \quad (26)$$

for one fixed measurable finite-output map f . Coloring row t by $t \bmod w$ satisfies the dependence contract with at most $d = w$ colors.

Proof. Two rows in the same color use disjoint innovation blocks. Every finite collection in that color is therefore mutually independent. The fixed map and the i.i.d. innovation law give the common row law. $\square$

This corollary covers a fixed categorical overlapping-window construction from i.i.d. innovations. It does not cover circular wraparound windows, a width selected from evaluation outcomes, a rolling-start sequence of repeated claims, or innovations with unspecified time dependence.

Definition 4.3 (Strong ℓ -dependence). Let $\ell \in \mathbb{N}_0$. An ordered sequence is strongly ℓ -dependent when

$$\sigma(Z_i : i \leq t) \text{ is independent of } \sigma(Z_i : i \geq t + \ell + 1) \quad (27)$$

for every t .

Coloring index t by $t \bmod (\ell + 1)$ separates consecutive members of each color by more than ℓ . Repeatedly separate the last row from the sigma-field generated by all preceding rows. This ordered induction proves mutual independence of all complete rows in each color. Pairwise lag independence alone does not prove this property. The common-law theorem still requires one common row law. If row laws differ, the drift theorem and its additional premises apply.

5 Nonidentical row laws and explicit drift

Assume $K \geq 2$. Let row i have law P_i , and keep the within-color mutual-independence contract. Define

$$\bar{P}_n = \frac{1}{n} \sum_{i=1}^n P_i. \quad (28)$$

Center the subset indicator for row i by $P_i(A)$. The Hölder–Hoeffding proof is unchanged.

Theorem 5.1 (Average-law and reference-law bounds). *For every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\hat{P}_n - \bar{P}_n\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, C_K \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (29)$$

For an explicit reference law $P_\star$, put

$$b_n = \left\|\bar{P}_n - P_\star\right\|_1. \quad (30)$$

Then

$$\mathbb{P}\left(\left\|\hat{P}_n - P_\star\right\|_1 \geq b_n + \varepsilon\right) \leq \min\left\{1, C_K \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (31)$$

Proof. Equation (29) follows from the row-specific centering. The triangle inequality gives

$$\left\|\hat{P}_n - P_\star\right\|_1 \leq \left\|\hat{P}_n - \bar{P}_n\right\|_1 + b_n, \quad (32)$$

which proves Equation (31). $\square$

If $\|P_i - P_\star\|_1 \leq \beta_i$, then convexity gives

$$b_n \leq \frac{1}{n} \sum_{i=1}^n \beta_i. \quad (33)$$

Under one fixed infinite row-law sequence and coloring with the same premises, the all-prefix reference-law envelope is

$$\mathbb{P}\left(\forall n \geq 1 : \|\hat{P}_n - P_\star\|_1 \leq \min\{2, b_n + D_n\}\right) \geq 1 - \alpha. \quad (34)$$

The bias term is necessary. Concentration about $\bar{P}_n$ does not identify a fixed scientific estimand.

6 A stronger local shared-exclusions modulus

Let p and q be two laws on the complete joint alphabet. Put

$$\delta = \|q - p\|_1, \quad p_{\min} = \min_{z \in \text{supp}(p)} p(z), \quad L = \log(1/p_{\min}). \quad (35)$$

Assume

$$\text{supp}(q) \subseteq \text{supp}(p), \quad 0 \leq \delta < 2p_{\min}. \quad (36)$$

Then the supports are equal. For every shared-exclusions event E that contains the keyed supported realization,

$$p(E) \geq p_{\min}, \quad |q(E) - p(E)| \leq \frac{\delta}{2}. \quad (37)$$

The factor one half follows from the variational identity; Lean proves the corresponding zero-total-mass event lemma.

All logarithms are natural, so all information values use nats. Fix a supported realization $z = (s_1, \dots, s_m, t)$ and a nonempty antichain α of nonempty source subsets. For a source subset a , the event $\{S_a = s_a\}$ fixes all source coordinates in a . Define

$$A_\alpha(s) = \bigcup_{a \in \alpha} \{S_a = s_a\}, \quad (38)$$

$$B_\alpha(s, t) = A_\alpha(s) \cap \{T = t\}, \quad (39)$$

$$C(t) = \{T = t\}. \quad (40)$$

Each event contains z . For a law Q whose three event masses are positive, define

$$c_\alpha^+(z; Q) = -\log Q(A_\alpha(s)), \quad (41)$$

$$c_\alpha^-(z; Q) = \log \frac{Q(C(t))}{Q(B_\alpha(s, t))}, \quad (42)$$

$$c_\alpha^{\text{sx}}(z; Q) = c_\alpha^+(z; Q) - c_\alpha^-(z; Q). \quad (43)$$

Let M be the fixed Möbius matrix for the selected redundancy lattice. For $u \in \{+, -, \text{sx}\}$, define the pointwise and averaged atoms

$$\pi_\alpha^u(z; Q) = \sum_\beta M_{\alpha\beta} c_\beta^u(z; Q), \quad (44)$$

$$\bar{\pi}_\alpha^u(Q) = \sum_{z \in \text{supp}(Q)} Q(z) \pi_\alpha^u(z; Q). \quad (45)$$

Define

$$\Lambda(\delta, p_{\min}) = -\log\left(1 - \frac{\delta}{2p_{\min}}\right). \quad (46)$$

Lemma 6.1 (One-log modulus). *Under Equation (36),*

$$|\log q(E) - \log p(E)| \leq \Lambda(\delta, p_{\min}). \quad (47)$$

Proof. Put $x = \delta/(2p_{\min}) < 1$. Event control gives

$$1 - x \leq \frac{q(E)}{p(E)} \leq 1 + x. \quad (48)$$

Monotonicity of the logarithm gives the lower side. For the upper side,

$$\log(1 + x) \leq x \leq -\log(1 - x). \quad (49)$$

The Lean artifact proves this argument over exact real numbers. $\square$

For one lattice node, the cumulative informative term contains one event logarithm. The misinformative term contains two. Their difference is the net term. Their errors are therefore at most

$$\Lambda, \quad 2\Lambda, \quad 3\Lambda. \quad (50)$$

Let M be the fixed Möbius matrix, and define the exact absolute row sum

$$r_\alpha = \sum_{\beta} |M_{\alpha\beta}|. \quad (51)$$

Lean proves the generic fixed-row inequality. Hence the pointwise informative, misinformative, and net atom errors are at most

$$r_\alpha \Lambda, \quad 2r_\alpha \Lambda, \quad 3r_\alpha \Lambda. \quad (52)$$

Each cumulative informative and misinformative value under p lies in $[0, L]$. Each cumulative net value lies in $[-L, L]$. A weight-change decomposition therefore gives the common-support averaged bounds

$$\begin{aligned} \text{informative: } & r_\alpha \left(\Lambda + \frac{L\delta}{2} \right), \\ \text{misinformative: } & r_\alpha \left(2\Lambda + \frac{L\delta}{2} \right), \\ \text{net: } & r_\alpha (3\Lambda + L\delta). \end{aligned} \quad (53)$$

Equations (52)–(53) are conservative generic Möbius-row baselines. They do not use the paper-defined nonnegativity of the informative and misinformative SxPID atoms. They are valid upper bounds, but this paper does not claim that they are sharp SxPID constants.

The informative and misinformative weight terms use $L\delta/2$. A half-factor for the net term does not follow from the generic cumulative-value range alone. The centered half-range tightening is new in **pid-rs**. It is project-defined validation, not a claim of scientific novelty or publication priority.

For completeness, the weight-change decomposition for one component is

$$\left| \sum_z q(z) \pi_\alpha^u(z; q) - \sum_z p(z) \pi_\alpha^u(z; p) \right| \leq \sum_z q(z) |\pi_\alpha^u(z; q) - \pi_\alpha^u(z; p)| \quad (54)$$

$$+ \left| \sum_z (q(z) - p(z)) \pi_\alpha^u(z; p) \right|. \quad (55)$$

For any finite real function f and signed weights h with $\sum_z h(z) = 0$, centering f at the midpoint of its range gives

$$\left| \sum_z h(z)f(z) \right| \leq \frac{\sum_z |h(z)|}{2} \left(\max_z f(z) - \min_z f(z) \right). \quad (56)$$

Here $h = q - p$ and $\sum_z |h(z)| = \delta$. At p , each cumulative informative and misinformative value is in $[0, L]$. Its Möbius atom therefore has oscillation at most $r_\alpha L$, which gives $r_\alpha L \delta / 2$. Each cumulative net value is in $[-L, L]$. Its atom has oscillation at most $2r_\alpha L$, which gives $r_\alpha L \delta$. The net calculation does not add separate informative and misinformative weight terms.

If $\delta \leq p_{\min}/2$, then $x \leq 1/4$. Lean also proves

$$\Lambda \leq \frac{x}{1-x} \leq \frac{4x}{3} = \frac{2\delta}{3p_{\min}}. \quad (57)$$

This improves the looser valid constants in the earlier finite-alphabet analysis. All statements in this section use exact real arithmetic.

7 Composition with SxPID

For common-law sampling, empirical support is a subset of population support almost surely. On the event in Theorem 4.1, the strict condition

$$D_n < 2p_{\min} \quad (58)$$

forces empirical and population support to agree. The realized error satisfies $\delta \leq D_n$, and $\Lambda(\delta, p_{\min})$ is nondecreasing in δ . Therefore, use D_n on the right-hand side of Equations (46)–(53). This gives simultaneous exact-real envelopes for every supported shared-exclusions cumulative term and every fixed Möbius atom.

A usable numerical envelope needs a justified lower bound on $p_{\min}$. The observed minimum frequency is not such a bound. A positive population cell can be absent from a finite sample.

For Theorem 5.1, also require every P_i to be supported inside $\text{supp}(P_\star)$. Define

$$p_{\min, \star} = \min_{z \in \text{supp}(P_\star)} P_\star(z). \quad (59)$$

If

$$b_n + D_n < 2p_{\min, \star}, \quad (60)$$

then the realized error is at most $b_n + D_n$. Monotonicity gives the same composition with $b_n + D_n$ on the right-hand side.

A frozen finite-output transform can be handled conditionally on an independent training artifact. Conditional on that artifact, the color map must be fixed. Every row must have the stated conditional law. The complete rows in every color class must be mutually independent. A random conditional support floor gives a conditional result unless a deterministic lower bound is available.

The displayed common-law envelope proves almost-sure exact-real finite-alphabet SxPID plug-in consistency under the sufficient condition $V_n \log(n)/n^2 \rightarrow 0$; fixed d is sufficient. The displayed drift envelope proves almost-sure exact-real reference-law SxPID consistency when this sufficient condition holds and $b_n \rightarrow 0$. These are not necessary conditions under a stronger sampling theorem. The result does not validate continuous $I_\cap^{\text{sx}}$, arbitrary overlapping windows, adaptive preprocessing on evaluation rows, unknown mixing rates, or binary64 asymptotics.

8 Counterexamples and invalidated routes

Invalid route	Counterexample	Required correction
Pairwise independence inside one color	Let U be uniform on $\mathbb{F}_2^4$. Use the 15 nonzero linear forms $a \cdot U$ as binary rows. Every pair is independent. All rows are zero with probability $1/16$. The empirical L^1 error is then one, while the false one-color bound is $2e^{-15/2} \approx 0.00111$.	Require mutual independence of all complete rows in each color.
Remove the color factor	Take m independent fair bits and copy each bit once into each of d colors. Each color is i.i.d., but only $m = n/d$ independent values remain.	Keep V_n , or keep the coarser factor d .
Infer convergence from one color per row	Set every row equal to one common fair bit. Singleton colors satisfy their local premise, but $V_n = n^2$ and the empirical law does not converge to the fair law.	This coloring cannot prove convergence; use a stronger coloring or another theorem.
Read the envelope condition as necessary	Let the rows be i.i.d. but assign each row a separate color. Then $V_n = n^2$, although the empirical law converges by the strong law.	State $V_n \log(n)/n^2 \rightarrow 0$ only as a sufficient condition for this envelope.
Data-adaptive coloring without the complete conditional contract	Set every row equal to one fair bit, then select one occupied color from that bit. Conditional rows are degenerate, but their conditional law is not the common unconditional law.	Require fixed colors, or prove the full conditional law and independence premises.
Substitute a mixing label for the color premise	Use a stationary fair two-state Markov chain with flip probability $\eta = 1/100$. Its transition dependence decays geometrically. For two rows, the empirical L^1 error is one with probability $1 - \eta = 0.99$, above the false one-color bound $2e^{-1}$.	Use a theorem with an explicit mixing coefficient and rate, or prove a valid independence coloring.
Pairwise lag independence as strong ℓ -dependence	The finite-field construction has pairwise independence and a joint failure.	Require the sigma-field definition. Residues modulo $\ell + 1$ then give valid colors by ordered induction.

Invalid route	Counterexample	Required correction
Halve the net weight term from the generic range alone	On two points, let $h = (-\delta/2, \delta/2)$ and a generic row value be $f = (-r_\alpha L, r_\alpha L)$. Then $ \sum hf = r_\alpha L\delta$. This row value is not shown to be realizable by an SxPID atom when $r_\alpha > 1$. It proves no SxPID sharpness claim.	Keep the generic range baseline unless a separate SxPID argument supplies a smaller range.
Allow $\delta = 2p_{\min}$	Move all mass from a least-probable supported cell. Its required logs become undefined.	Keep $\delta < 2p_{\min}$ strict.
Control only separate marginals	Set $S_1 = S_2 = X$. For $0 < \rho < 1$, at the supported realization $(X, T) = (0, 0)$, use fair binary X, T marginals with diagonal cells $(1 + \rho)/4$ and off-diagonal cells $(1 - \rho)/4$, then replace ρ by $-\rho$. Marginals stay fixed, but pointwise redundancy changes by $\log((1 + \rho)/(1 - \rho))$.	Control the complete joint law.
Allow new support with a linear average bound	Let $p(0, 0, 0) = 1$, and let $q(0, 0, 0) = 1 - \varepsilon$, $q(1, 1, 1) = \varepsilon$. Then $\ q - p\ _1 = 2\varepsilon$, while the averaged redundancy is $-(1 - \varepsilon)\log(1 - \varepsilon) - \varepsilon\log\varepsilon$.	Require common support or use a nonlinear boundary modulus.
Use observed minimum frequency as a population floor	An unobserved positive cell has empirical frequency zero.	Use an external floor or a separate simultaneous lower-confidence result.

The executable challenge suite enumerates the finite-field pairwise-independence construction, copied-color construction, singleton-color construction, and adaptive-color construction. It also checks a support-deletion boundary, an unspecified-mixing construction, a generic net-weight range extremizer, a univariate-marginal-only construction, and a new-support construction. The remaining checks cover the telescoping allocation, class-size inequalities, high-precision one-log cases, and all displayed bounds on three committed two-source law pairs.

The fixed-window fixture uses independent fair innovations U_t, V_t and

$$S_{1,t} = U_t \text{ OR } U_{t+1}, \quad S_{2,t} = V_t \text{ OR } V_{t+1}, \quad T_t = S_{1,t} \text{ XOR } S_{2,t}. \quad (61)$$

Even and odd starts are the two colors. Its exact joint count weights are $1 : 3 : 3 : 9$. The high-precision Decimal oracle supplies expected logarithmic values. For stored logarithmic constants and bounds, the Rust reconstruction tolerance is $32\varepsilon_{\text{mach}} \max(1, |x_{\text{Rust}}|, |x_{\text{oracle}}|)$. For categorical estimator outputs, the absolute comparison ceiling is $32\varepsilon_{\text{mach}}$ nats, where $\varepsilon_{\text{mach}} = 2^{-52}$ for binary64. The test also checks that row reversal gives bit-identical output.

9 Formal and executable evidence boundary

The Lean project is pinned to one toolchain and one exact dependency manifest. The repository checker rejects admitted proof placeholders, builds the project, and replays its declarations with Lean’s kernel checker.

The new Lean module proves:

- the event-mass $\delta/2$ lemma and exact attainment by the positive-coordinate event;
- positivity under the strict support margin;
- the one-log modulus, its monotonicity, and its rational simplification;
- the effective-color proxy bounds and the normalized factor range;
- the generic absolute linear-row bound;
- generic finite weighted-average perturbation bounds, including the centered half-range form;
- the exact unit-scale finite telescoping allocation; and
- the algebraic radius-exponent cancellation.

Lean does not encode random variables or conditional laws. It does not encode generalized Hölder, Hoeffding’s lemma, the Chernoff argument, or the probability union bound. The drift result and fixed-window independence argument also remain outside Lean. Lean does not encode SxPID events or the redundancy lattice. The identification of its generic lemmas with SxPID atoms also remains outside Lean. Rust refinement and binary64 rounding remain outside Lean. The probability theorem is a prose proof. Lean machine-checks only the deterministic obligations listed above. The Python suite uses exact rational arithmetic for finite probability and count identities, plus 100-digit Decimal arithmetic for logarithms. Transcendental logarithms are not exact rational values. From each local count-table pair, the Rust test independently reconstructs the two empirical laws, δ , $p_{\min}$, Λ , L , and every stored node bound before it checks the implementation outputs. It uses the scale-aware reconstruction tolerance and the absolute categorical-output ceiling declared above. These checks are bounded executable evidence. They are not a general proof, a global binary64 bound, or external review.

10 Ecosystem use boundary

Consumer	Result supplied here	Still required
Prisoma	Categorical SxPID law control after a frozen finite map when held-out rows meet the color contract	Concrete row law, justified support floor, independent transform receipt, and a valid window coloring
Galadriel	Dependence-aware categorical alternative to an i.i.d. plug-in claim	Removal of same-window fitting and generic jitter routes; explicit observation and coloring contracts
Haldir	Mathematical target for block or residue-class designs	Direct dependency, corrected atom interpretation, and proof of strong ℓ -dependence or another valid coloring
Crebain	Finite-alphabet validation component for a future integration	Exact observation mapping, run-log contract, direct dependency, and explicit scientific estimand

This document does not assert that a consumer meets these premises. Integration remains `not_claimed` until the consumer records and checks them.

11 Reproduction

From the repository root, run

```
python3 scripts/check-lean-finite-convergence.py
python3 scripts/generate-dependency-colored-sxpid-oracle.py
scripts/check-dependency-colored-sxpid-pdf.sh
```

The PDF checker rebuilds this source with a fixed source-date epoch, rejects LaTeX diagnostics, and compares the exact PDF bytes. The Lean checker binds the toolchain, manifest, source set, dependency revisions, dependency origins, and clean checkout state. It removes ambient Git routing and global or system configuration while retaining each dependency checkout's local origin configuration for verification.

12 Conclusion

The result supplies a dependence-aware finite-sample path from a declared finite-row coloring to exact-real categorical shared-exclusions PID bounds. Its displayed envelope proves almost-sure exact-real consistency under the sufficient condition $V_n \log(n)/n^2 \rightarrow 0$. The corresponding reference-law result also uses $b_n \rightarrow 0$. These are not necessary conditions under a stronger sampling theorem. The result separates four objects that must remain separate: sampling fluctuation, deterministic drift, support margin, and program arithmetic. The color-size proxy improves the proof constant for unbalanced classes. The telescoping allocation gives a simple time-uniform statement. The local logarithmic modulus gives a direct law-to-atom transfer once common support is justified. Every weaker route listed in Section 8 is retained as invalidated work.

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